Nitin S Katagihallimath

AI Engineer | Python | LLM Agents & RAG
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Summary

Electrical and Electronics Engineering graduate with hands-on Python experience building LLM-powered agents and Retrieval-Augmented Generation (RAG) systems. Built FastAPI services, integrated LLM APIs, embeddings and vector search (FAISS), and orchestrated agent workflows with LangGraph. Tested an AI agent at multiple levels (unit, live-service and end-to-end), and trained and evaluated deep learning models in PyTorch.

Technical Skills

- **Core**: Python, SQL, Git, GitHub, Pytest
- **GenAI & RAG**: LLM Agents (Tool Calling), RAG Pipelines, Embeddings, Vector Search (FAISS), LangChain, LangGraph, Prompt Design, Gemini, Groq, Hugging Face, Ollama
- **APIs & Automation**: FastAPI (REST APIs), JWT Authentication, Docker, n8n
- **ML & Data**: PyTorch, Pandas, NumPy, OpenCV, Model Evaluation (Accuracy, SMAPE), Grad-CAM Explainability

Work Experience

AI/ML Engineer Intern | CAIR Lab, Hubballi — Certificate | *Jan 2026 – May 2026*

- Fine-tuned DenseNet169 on a curated **12,000-image** oral lesion dataset for 4-class classification, achieving **94.5% accuracy**;used Grad-CAM to check which image regions drove each prediction.
- Developed and deployed an AI application integrating a FastAPI backend, LangGraph, and React for image uploads, model-based risk scoring, and clinical report generation.

Healthcare Chatbot Developer | Freelance, Remote — *May 2026 – Present*

- Built a **4-language** CLD and sarcopenia patient-education chatbot with **RAG** (FAISS + Gemini), assessments, and PDF reports; pending client handover after ethical clearance.
- Added JWT authentication and Supabase report management.

Projects

Open-Source Contribution Copilot – ASI:One Agent Challenge — GitHub

- Built an AI agent to discover open-source GitHub issues and automate contribution steps, including repository forking, claim comments, and branch creation using the GitHub API.
- Registered the agent on Agentverse and enabled discovery through ASI:One using the Agent Chat Protocol. *Stack: Agentverse, ASI:One, Agent Chat Protocol, GitHub API*
- Tested across layers: **113 pytest unit tests**, live checks against the real GitHub API and LLM, and an end-to-end harness over the chat protocol; live runs exposed bugs (rate limits, empty-result handling) that unit tests missed.

CoDe-DuINo – AI Arduino Development Assistant — GitHub | Live Demo

- Built an **LLM agent** that generates, compiles, and uploads Arduino code via tool calling, with voice input, pin mapping, and live serial logs.
- Cut hardware setup time **from hours to minutes**. *Stack: LangGraph, FastAPI, Groq API, arduino-cli, Render*

Multimodal Product Price Prediction – Amazon ML Hackathon 2025 — GitHub

- Combined DistilBERT text embeddings and ResNet50 image embeddings to develop a multimodal product price prediction pipeline.
- Used an XGBoost regressor for price prediction, achieving a reported **SMAPE of 61.8**. *Stack: Python, DistilBERT, ResNet50, XGBoost*

Education

KLE Technological University, Hubballi – B.E. in Electrical and Electronics Engineering, CGPA 8.10 — *[June 2026]*

Prism PU Science College, Dharwad – Pre-University (PCMB) : 89.16% — *[Apr 2022]*

Shanti Sadan High School, Dharwad – SSLC : 89.92% — *[Jul 2020]*

Certifications

- **NLP Specialization – DeepLearning.AI (Coursera)**Link — *[Oct 2025]*
- **FastAPI for Machine Learning** Link — *[JUL 2026]*

Achievements

- **Research paper accepted at ICMLAS 2025** – Text-to-Image GAN (TextControlGAN) [cert]
- **1st place – Nivista 3.0 Project Exhibition** for Solar-Assisted Backup Propulsion System for EVs [cert]

Languages & Interests

- **Languages**: English · Kannada · Hindi
- **Interests**: Sketching · Table Tennis · Video editing · Art-craft · Cooking